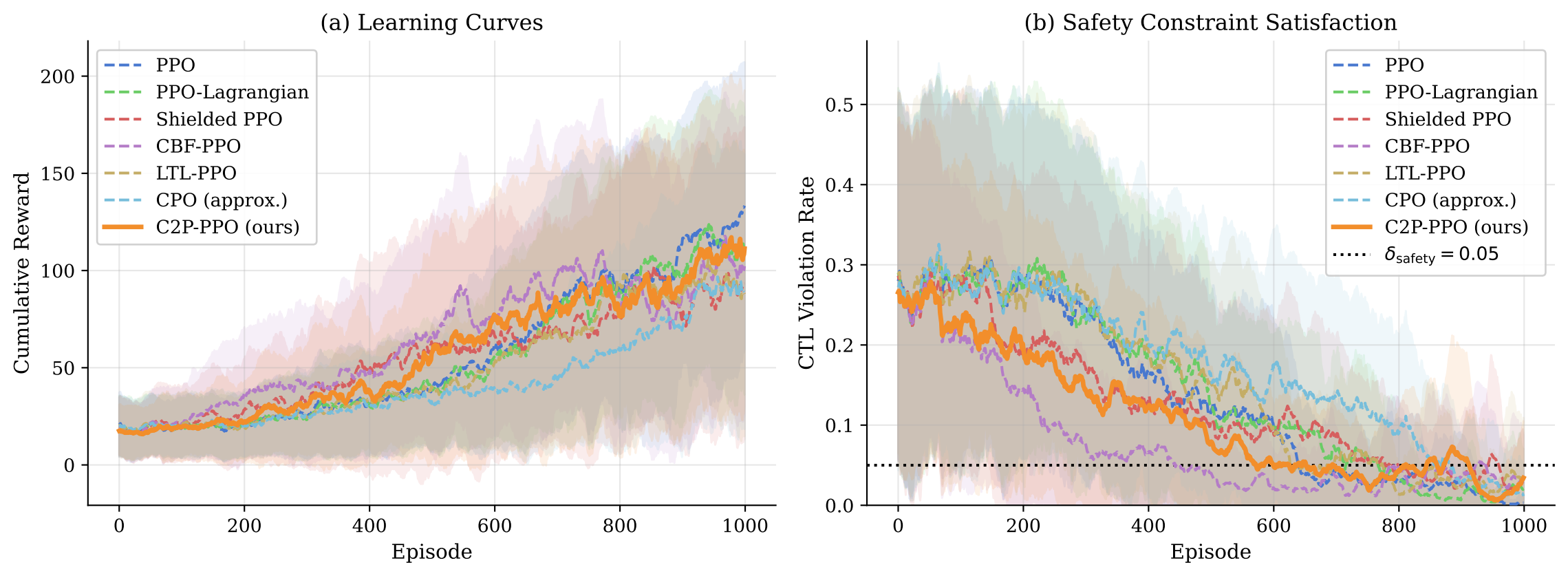

(a) Learning Curves
Cumulative Reward
Episode
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)
(b) Safety Constraint Satisfaction
CTL Violation Rate
Episode
$\delta_{\text{safety}} = 0.05$